

Multi-Build Comparative Analysis: SLM Production of Ti6Al4V Femoral Stems

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Prepared by: Advanced Manufacturing Engineering Team

Scope: Analysis of three consecutive build jobs for medical implant production

Executive Summary

This report provides a detailed comparative analysis of three selective laser melting (SLM) production runs for Ti6Al4V femoral stems conducted between Q2 2024 and Q3 2024. The analysis focuses on production efficiency, resource utilization, and process stability across build cycles. Key findings indicate consistent product quality with variations in material and energy efficiency. The current build demonstrates improved gas management and slightly reduced energy consumption compared to previous runs, though opportunities remain for optimizing powder utilization.

The femoral stem production represents a critical medical component requiring precise dimensional accuracy and material properties. All builds maintained the target output of 20 stems per job with identical geometric specifications. This analysis examines the operational parameters and resource flows to identify trends and improvement opportunities in our additive manufacturing process.

1. Introduction

The manufacturing of orthopedic implants via SLM technology has become increasingly important for producing patient-specific components with complex geometries. This analysis covers three production builds of Ti6Al4V femoral stems, with particular focus on the most recent build completed in Q3 2024 (designated as Current Build). The comparative approach allows for tracking process evolution and identifying areas for continuous improvement.

Production of medical implants demands strict adherence to quality standards while maintaining economic viability. This report examines the operational data from multiple perspectives to provide insights for production planning and process optimization.

Build Series Overview: - Build 1 (Q2 2024): Initial production run with standard parameters - Build 2 (Q3 2024 - Previous): Process-optimized run - Current Build (Q3 2024 - Latest): Most recent production with refined parameters

All builds utilized the same SLM machine platform (Concept Laser M2) with identical build chamber dimensions and laser configuration.

2. Methodology and Build Parameters

2.1 Equipment and Material Specifications

The production utilized a standardized SLM platform with the following fixed parameters across all builds:

- **Machine:** Concept Laser M2
- **Build Chamber Volume:** 250 × 250 × 280 mm
- **Laser Type:** Fiber laser, 400 W nominal power
- **Layer Thickness:** 30 µm
- **Scan Speed:** 800 mm/s
- **Atmosphere Control:** High-purity argon (>99.999%)
- **Powder Specification:** Ti6Al4V ELI, plasma atomized, 15-45 µm particle size distribution

2.2 Build Configuration

All three builds employed identical nesting strategies to maximize build chamber utilization while maintaining adequate part spacing for thermal management. The support structure design remained consistent across builds, utilizing block-type supports with 0.8 mm contact diameter to facilitate post-process removal.

Key Build Parameters:

Parameter	Build 1	Build 2	Current Build
Number of Stems	20	20	20
Total Build Time (hours)	63.2	62.1	61.35
Average Machine Power (kW)	2.45	2.42	2.4
Laser Power Setting (W)	400	400	400
Chamber Pre-heat Temperature (°C)	180	170	165

The gradual reduction in build time and machine power consumption reflects ongoing process optimization efforts, particularly in scan path strategies and thermal management.

3. Material Utilization Analysis

3.1 Powder Consumption and Yield

Material efficiency represents a significant cost factor in SLM production, particularly with premium aerospace-grade Ti6Al4V powder. The analysis reveals consistent powder usage patterns with minor variations attributable to machine calibration and recycling protocols.

Powder Management Data:

Metric	Build 1	Build 2	Current Build
Fresh Powder Input (kg)	21.15	20.95	20.83
Final Product Mass (kg)	1.77	1.77	1.77
Unmelted Powder Recovered (kg)	19.25	19.08	18.99
Material Utilization Efficiency (%)	8.37	8.45	8.50

Material utilization efficiency calculated as: $(\text{Product Mass} / \text{Fresh Powder Input}) \times 100$

The current build shows a slight improvement in powder efficiency, though the overall utilization rate remains consistent with industry expectations for complex medical components with substantial support structures.

3.2 Powder Recycling and Management

The unmelted powder recovery system demonstrated consistent performance across builds. All recovered powder underwent standard sieving and characterization procedures before being reintroduced to the powder management system at a 50% refresh rate with virgin material.

Powder Balance Analysis (Current Build): - Fresh powder input: 20.83 kg - Powder accounted in products and waste: 1.77 kg (stems) + 18.99 kg (recyclable) + 0.019 kg (support waste) + 0.0208 kg (filter waste) = 20.7998 kg - System loss (handling, measurement variance): 0.0302 kg (0.15%)

The minimal system loss falls within acceptable tolerances for industrial powder handling operations.

4. Energy Consumption Profile

4.1 Electricity Utilization

Energy monitoring during the build process captured consumption from all subsystems including lasers, motors, cooling, and control systems. The data shows a positive trend toward reduced energy usage per build.

Energy Consumption Comparison:

Energy Metric	Build 1	Build 2	Current Build
Total Electricity (kWh)	154.8	150.3	147.26
Energy per Stem (kWh/stem)	7.74	7.52	7.36
Specific Energy (kWh/kg product)	87.5	84.9	83.2

The current build achieved a 4.9% reduction in total energy consumption compared to Build 1, primarily through optimized laser scanning strategies and improved thermal management reducing cooling requirements.

4.2 Power Demand Characteristics

Analysis of power consumption patterns revealed consistent machine operation with the expected profile: initial high consumption during chamber preparation, stable operation during the main build phase, and reduced power during cooldown.

Current Build Power Profile: - Chamber preparation: 3.1 kW average (2 hours) - Main build phase: 2.4 kW average (58.35 hours) - System cooldown: 1.2 kW average (1 hour)

The maintained average power consumption of 2.4 kW during the build phase indicates stable process conditions and consistent laser operation.

5. Process Gas Management

5.1 Argon Consumption Analysis

Inert gas usage represents both a operational cost and process stability factor. The SLM process utilizes argon for both initial chamber flooding and continuous atmosphere maintenance during the build phase.

Gas Consumption Data:

Gas Usage Phase	Build 1	Build 2	Current Build
Chamber Flooding (kg)	3.25	3.12	3.03
Building Phase (kg)	26.8	26.2	25.94
Total Argon (kg)	30.05	29.32	28.97
Gas per Stem (kg/stem)	1.50	1.47	1.45

The current build shows continued improvement in gas efficiency, achieving a 3.6% reduction compared to Build 1. This improvement stems from enhanced seal integrity and optimized gas flow rates that maintain the required oxygen levels (<1000 ppm) while minimizing consumption.

5.2 Gas Purity and Process Atmosphere

All builds maintained oxygen levels below 500 ppm throughout the process, well within the specification limit of 1000 ppm for Ti6Al4V processing. The consistent gas purity contributed to the absence of observable oxidation in all produced components.

Current Build Atmosphere Control: - Initial oxygen level after flooding: 120 ppm - Average oxygen during build: 280 ppm - Peak oxygen level: 450 ppm (during powder recoating) - Final oxygen level: 310 ppm

The stable atmosphere control ensured consistent melt pool behavior and minimal spatter generation throughout the 61.35-hour build cycle.

6. Waste Stream Management

6.1 Solid Waste Characterization

The SLM process generates minimal solid waste, primarily consisting of support structures removed during post-processing and fine powder captured by the filtration system.

Waste Generation Comparison:

Waste Stream	Build 1	Build 2	Current Build
Support Structures & Losses (kg)	0.022	0.020	0.019
Filter-Captured Powder (kg)	0.0235	0.0215	0.0208
Total Waste (kg)	0.0455	0.0415	0.0398
Waste per Stem (g/stem)	2.28	2.08	1.99

The downward trend in waste generation reflects improvements in support structure design and reduced spattering during the melting process. The current build achieved a 12.5% reduction in total waste compared to Build 1.

6.2 Waste Disposition and Recycling

All waste streams are managed according to established protocols:

Support Structures and Minor Losses (0.019 kg in Current Build): - Segregated as titanium scrap - Sent to certified metal recycling facility - Typically achieves >95% material recovery rate

Filter-Captured Powder (0.0208 kg in Current Build): - Comprises fine particles (<10 µm) and agglomerates - Designated for landfill disposal due to potential contamination and poor flow characteristics - Represents <0.1% of total powder input

The high recovery rate of unmelted powder (18.99 kg in Current Build) significantly reduces raw material requirements for subsequent builds, with the recycled powder maintaining suitable characteristics for medical component production.

7. Production Efficiency Metrics

7.1 Overall Process Efficiency

Combining the various resource streams provides a comprehensive view of production efficiency. The analysis focuses on the relationship between input resources and final product output.

Efficiency Indicators:

Efficiency Metric	Build 1	Build 2	Current Build
Build Rate (stems/day)	7.59	7.73	7.82
Machine Utilization (%)	87.2	88.5	89.1
Powder-to-Product Ratio	11.95:1	11.84:1	11.77:1
Energy Intensity (kWh/stem)	7.74	7.52	7.36

The steady improvement across most efficiency metrics demonstrates the effectiveness of continuous process refinement. The current build shows the highest machine utilization rate at 89.1%, indicating reduced non-productive time during chamber preparation and cooldown.

7.2 Cost Implications

While detailed cost analysis falls outside this report’s scope, the resource efficiency trends directly impact production economics:

- **Powder Cost:** The 1.5% reduction in fresh powder usage from Build 1 to Current Build translates to approximately \$150 savings per build at current Ti6Al4V pricing
- **Energy Cost:** Reduced electricity consumption represents approximately \$7 savings per build at industrial rates
- **Gas Cost:** Argon consumption reduction equates to roughly \$25 savings per build

These incremental improvements contribute to enhanced competitiveness for medical implant manufacturing.

8. Root Cause Analysis of Variations

8.1 Performance Drivers

The comparative analysis reveals several factors contributing to the observed performance variations:

Build Time Reduction (61.35 hours in Current Build vs. 63.2 hours in Build 1): - Optimized scan path strategies reduced non-melting travel time - Improved recoater blade performance enabled faster powder deposition - Enhanced thermal management reduced inter-layer cooling requirements

Energy Efficiency Improvement (147.26 kWh in Current Build vs. 154.8 kWh in Build 1): - Lower average machine power (2.4 kW vs. 2.45 kW) through optimized subsystem operation - Reduced build duration directly decreased energy consumption - Improved cooling system efficiency

Material Efficiency Gains: - Enhanced parameter sets reduced spatter generation - Improved powder recycling protocols minimized handling losses - Optimized support structures decreased sacrificial material

8.2 Process Stability Assessment

All three builds demonstrated excellent process stability with no build failures or significant interruptions. The consistency in final product mass (1.77 kg across all builds) indicates robust process control and reproducible melting behavior.

Quality Metrics Consistency: - Dimensional accuracy: All stems within ± 0.1 mm specification - Density: $>99.7\%$ measured via Archimedes method - Surface roughness: Ra 12-15 μm as-built condition - Mechanical properties: Yield strength 950-1010 MPa, meeting ASTM F136 specifications

9. Recommendations for Process Optimization

Based on the comparative analysis, the following recommendations are proposed for future builds:

9.1 Immediate Actions (Next 1-2 Builds)

- Powder Management:** Implement enhanced sieving protocols to potentially increase recyclable powder yield by 0.5-1.0%
- Gas Flow Optimization:** Conduct controlled experiments to determine minimum argon flow rates that maintain atmosphere purity, targeting 5% additional reduction
- Support Structure Refinement:** Further optimize support design to reduce material in low-stress regions, potentially saving 0.005-0.010 kg per build

9.2 Medium-term Initiatives (Next Quarter)

- Energy Recovery:** Evaluate feasibility of waste heat recovery from cooling systems for facility heating
- Build Packing Optimization:** Investigate alternative nesting configurations to potentially increase stem count to 22 per build without compromising quality
- Predictive Maintenance:** Enhance monitoring of filter system to extend service life and reduce waste powder generation

9.3 Strategic Considerations

- Powder Recycling Economics:** Assess the cost-benefit of advanced powder characterization equipment to potentially increase refresh ratios
 - Alternative Gas Sources:** Investigate nitrogen-argon mixtures for non-critical process phases where oxygen sensitivity is reduced
 - Machine Upgrade Planning:** Evaluate next-generation SLM systems with improved energy efficiency and faster build rates for future capital planning
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10. Conclusions

The multi-build comparative analysis demonstrates a consistently improving SLM process for Ti6Al4V femoral stem production. The Current Build achieved the highest efficiency across most metrics while maintaining product quality standards.

Key achievements include: - 4.9% reduction in energy consumption compared to Build 1 - 3.6% reduction in argon usage - 12.5% reduction in waste generation - Maintained product quality and dimensional accuracy

The incremental improvements reflect successful implementation of previous recommendations and ongoing process refinement. The data provides a solid foundation for continued optimization efforts and supports the economic viability of SLM for medical implant manufacturing.

The process demonstrates maturity and stability suitable for serial production, with consistent output of 20 high-quality femoral stems per build within the expected resource envelopes.

Appendix A: Data Collection Methodology

All data collected via calibrated monitoring systems: - Powder mass: Mettler Toledo industrial scales (± 0.001 kg accuracy) - Gas flow: Bronkhorst mass flow meters ($\pm 1\%$ accuracy) - Electricity: Schneider PowerLogic meters ($\pm 0.5\%$ accuracy) - Build parameters: Machine integrated monitoring system

Appendix B: Build Identification

- Build 1: Job ID SLM-MED-024-061 (Completed June 12, 2024)
 - Build 2: Job ID SLM-MED-024-078 (Completed August 8, 2024)
 - Current Build: Job ID SLM-MED-024-095 (Completed October 15, 2024)
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